

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 18: Aerodrome Design & OLS Surfaces

Syllabus Source: CAE Oxford (pp. 375-398) | Pass Target: $\geq 90\%$ AviationExam

1. AERODROME REFERENCE CODE (ANNEX 14)

Annex 14 classifies aerodrome facilities using a two-element reference code based on critical aircraft design:

Code No.	Airplane Reference Field Length	Code Letter	Wingspan & Outer Main Gear Wheel Span
Code 1	Reference Field Length < 800 m	Code A	Wingspan < 15 m (Gear span < 4.5 m)
Code 2	Field Length 800 m to < 1,200 m	Code B	Wingspan 15 m to < 24 m (Gear span 4.5 to < 6 m)
Code 3	Field Length 1,200 m to < 1,800 m	Code C	Wingspan 24 m to < 36 m (Gear span 6 to < 9 m) [A320, B737]
Code 4	Field Length $\geq 1,800$ m	Code D	Wingspan 36 m to < 52 m (Gear span 9 to < 14 m) [B767]
--	--	Code E	Wingspan 52 m to < 65 m (Gear span 9 to < 14 m) [B777, A350]
--	--	Code F	Wingspan 65 m to < 80 m (Gear span 14 to < 16 m) [A380, B747-8]

2. RUNWAY STRIPS, CLEARWAYS & STOPWAYS

- **Runway Strip:** A defined area including runway and stopway intended: 1) To reduce damage if aircraft veers off runway, 2) To protect aircraft flying over during take-off or landing. For Code 3/4 precision runways, extends 60 m beyond each runway end, with a total width of at least 280 m (140 m each side of centerline) or 300 m (150 m each side).
- **Clearway:** Rectangular area on ground or water under control of aerodrome, selected or prepared as suitable area over which an aeroplane may make a portion of its initial climb to a specified height. Maximum length shall not exceed HALF THE LENGTH OF TORA (Clearway max = $0.5 \times \text{TORA}$). Upward slope max 1.25%.
- **Stopway:** Defined rectangular area on ground at end of take-off run available, prepared as suitable area in which an aircraft can be stopped in the event of an abandoned take-off. Width equal to runway.

3. DECLARED DISTANCES (TORA, TODA, ASDA, LDA)

Declared distances are the operational lengths available for aircraft performance calculations:

Declared Distance	Calculation Formula & Operational Definition
TORA (Take-Off Run Available)	Length of runway declared available and suitable for the ground run of an aircraft taking off.
TODA (Take-Off Distance Available)	TORA + Clearway. (The length of TORA plus length of clearway, if provided).
ASDA (Accelerate-Stop Distance Available)	TORA + Stopway. (The length of TORA plus length of stopway, if provided).

LDA (Landing Distance Available)

Length of runway declared available and suitable for the ground run of an aeroplane landing. Measured from threshold.

[!] EXAM TRAP (AVIATIONEXAM) - Displaced Threshold & Practical Calculation Exercise

When a runway threshold is **DISPLACED** (moved down the runway): - Landing Distance Available (LDA) is **REDUCED** by the displacement distance. - Take-Off Run Available (TORA), TODA, and ASDA are **NOT** reduced for take-off in that direction. - The area preceding a displaced threshold is suitable for take-off and roll, but **NEVER** for landing.

4. RUNWAY END SAFETY AREA (RESA)

An area symmetrical about extended runway centerline intended to reduce risk of damage to aircraft undershooting or overrunning runway. For Code 3 and 4 runways: minimum length is 90 m (recommended 240 m); width must be at least **TWICE** the width of the associated runway.

5. OBSTACLE LIMITATION SURFACES (OLS)

OLS Surface	Dimensions & Operational Protection Objective
Approach Surface	Inclined plane preceding threshold. Slope varies from 2.0% (Code 4 precision) to 3.3%.
Transitional Surface	Complex surface sloping upward and outward from runway strip edges to the inner horizontal surface. Standard slope: 14.3% (1:7) for Code 3/4 runways.
Inner Horizontal Surface	Circular horizontal plane located 45 m above aerodrome elevation (radius 4,000 m for Code 4). Protects visual maneuvering area.
Conical Surface	Surface sloping upward and outward from perimeter of inner horizontal surface at a 5% slope to a specified height (100 m for Code 4).
Balked Landing Surface	Inclined plane located after threshold at a specified distance protecting aircraft executing a missed approach at very low altitude.

6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the maximum allowable length of a clearway relative to TORA?**

- [A]: 25% of TORA
- [B]: 50% of TORA
- [C]: Equal to TORA
- [D]: Unlimited provided slope is < 1.25%

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under ICAO Annex 14, the length of a clearway shall not exceed half the length of the take-off run available (0.5 x TORA).

Q2: At what height above aerodrome elevation is the Inner Horizontal Surface established?

- [A]: 30 m
- [B]: 45 m
- [C]: 60 m
- [D]: 100 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Inner Horizontal Surface is located 45 m above the aerodrome elevation datum.

Q3: What is the standard minimum length of a Runway End Safety Area (RESA) for Code 4 runways?

- [A]: 60 m
- [B]: 90 m (recommended 240 m)
- [C]: 120 m
- [D]: 300 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Standard minimum RESA length is 90 m, with 240 m recommended by ICAO.

Q4: Given: Runway length = 2,500 m, Displaced threshold = 300 m, Clearway = 400 m, Stopway = 200 m. What is the LDA?

- [A]: 2,500 m
- [B]: 2,200 m
- [C]: 2,700 m
- [D]: 2,900 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. LDA is measured from the threshold to runway end: 2,500 m - 300 m displacement = 2,200 m.

Q5: What is the slope of the transitional surface for a Code 4 precision approach runway?

- [A]: 5.0%
- [B]: 10.0%
- [C]: 14.3% (1:7)
- [D]: 20.0%

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. The transitional surface slope for Code 4 runways is 14.3% (or 1:7).